

LECTURE NOTES ON ALTERNATIVE ENERGY

The growing global demand for energy, coupled with the environmental consequences of fossil fuel consumption, has intensified the search for cleaner and more sustainable alternatives. Alternative energy refers to energy sources that are renewable, environmentally friendly, and capable of reducing dependence on conventional petroleum-based fuels. These include solar, wind, biomass, geothermal, hydropower, hydrogen, and emerging technologies such as gas-to-liquids (GTL) and biofuels. Unlike fossil fuels, which are finite and contribute significantly to greenhouse gas emissions, alternative energy sources offer long-term availability and reduced environmental impact, making them essential to climate change mitigation strategies.

Over the past decade, rapid technological advancements have increased the efficiency, reliability, and affordability of alternative energy systems. Countries around the world are integrating these resources into their energy mix to enhance energy security, diversify supply, and support sustainable economic growth. In regions like Africa, where vast renewable resources remain untapped, alternative energy holds particular promise for expanding electricity access, stimulating industrial development, and reducing the environmental footprint of energy production. As the world transitions toward a low-carbon future, the role of alternative energy becomes increasingly critical, providing a pathway to reduce emissions while meeting the energy needs of a growing global population.

Examples of Alternative fuels

- Bio-diesels
- Methanol
- Refused-derived fuels
- Chemically stored electricity (Fuel cells and Batteries)
- Hydrogen
- Non-fossil methane
- Non-fossil Natural gas
- Vegetable oil
- Biomass

The global energy landscape is undergoing a significant transformation as nations confront the environmental, economic, and sustainability challenges associated with conventional fossil fuels. Rapid industrialization, population growth, and the rising demand for energy have intensified pressures on existing fuel resources. At the same time, the impacts of climate change, air pollution, and resource depletion have highlighted the urgent need for cleaner and more reliable energy alternatives. Alternative fuels, derived from renewable or low-carbon sources, offer a promising pathway toward achieving a more sustainable and resilient energy future.

Need for alternative fuels

- Traditional sources of fuels will become depleted, because they are non-renewable
- Non-environmentalists also concur with the opinion that most oil fields (situated in the Middle East) in the world are associated with problems – both political and economic.

- Determining a new method or solution with respect to finding different countries to create new fuels would reduce the unrest and conflict resulting from the world's dependence on fuel supply.
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Why Alternative Energy Is Gaining So Much Attention?

In recent years, alternative energy has moved to the center of global discussions on sustainability, economic development, and environmental protection. This growing attention is driven by a combination of scientific, economic, and social factors that highlight the limitations of traditional fossil fuels and the advantages of cleaner, renewable energy sources. As the world faces increasing climate-related challenges such as extreme weather events, rising temperatures, and environmental degradation, the urgency to transition toward more sustainable energy systems has never been greater.

One of the primary reasons for the heightened focus on alternative energy is the pressing need to reduce greenhouse gas emissions. Fossil fuels remain the largest contributors to carbon dioxide emissions, which are directly linked to global warming. Renewable energy sources such as solar, wind, hydro, geothermal, hydrogen, and bioenergy offer cleaner pathways that significantly lower emissions, making them essential components of global climate action strategies. International agreements like the Paris Accord have further accelerated the shift toward low-carbon energy solutions.

Another major factor driving interest in alternative energy is energy security. Many countries depend heavily on imported oil and gas, exposing them to price volatility, political instability, and supply disruptions. Renewable energy, on the other hand, is locally available, abundant, and less vulnerable to geopolitical uncertainties. By diversifying their energy mix, nations can strengthen their independence and stabilize their economies against fluctuating global fuel prices.

Technological advancements have also made alternative energy more attractive. Improvements in battery storage, solar panel efficiency, wind turbine design, and hydrogen production have lowered costs and enhanced performance. As a result, renewable energy systems are becoming increasingly competitive with fossil fuels, especially in power generation and transportation. This shift not only reduces costs for consumers but also opens new opportunities for innovation, investment, and job

creation, also, social awareness and environmental advocacy are further strengthening the global push toward alternative energy. Communities, researchers, and governments are increasingly conscious of the long-term impacts of pollution, deforestation, and resource depletion. Public support for clean energy policies and sustainable practices has grown significantly, influencing national agendas and shaping global energy transitions.

In essence, the growing attention to alternative energy is driven by a convergence of environmental necessity, economic opportunity, technological progress, and global policy commitments. Together, these factors make alternative energy not just an option, but an imperative for a sustainable and resilient future.

The idea that *“trash for one person is a treasure for another”* captures the essence of how modern bioenergy systems transform waste and agricultural residues into valuable resources. Biofuels, bioproducts, and biopower illustrate this principle by converting materials that would otherwise be discarded, such as crop residues, animal waste, used cooking oil, wood chips, and municipal organic waste,

into useful energy products. Instead of becoming environmental pollutants, these materials are upgraded into fuels like biodiesel, bioethanol, and biogas, or used to produce electricity and heat. This not only reduces waste disposal challenges but also helps create a circular economy where resources are reused and recycled efficiently.

Furthermore, the production of biofuels from locally grown crops brings significant socioeconomic benefits. When farmers cultivate feedstock such as maize, sugarcane, cassava, oil palm, or jatropha for biofuel production, they gain additional income streams beyond traditional food markets. Local processing industries also create jobs, stimulate rural economies, and encourage infrastructural development. By sourcing raw materials from local farmers, biofuel industries help reward their hard work, promote agricultural productivity, and reduce the economic migration of rural populations. In this way, biofuels not only support clean energy goals but also empower local communities and strengthen regional economies.

FACTORS TO CONSIDER WHILE DECIDING CHOICE OF ALTERNATIVE FUEL

1. **Energy Efficiency and Performance**

- The fuel's heating value, power output, and suitability for the intended application (transportation, power generation, industrial use).

2. **Availability of Feedstock**

- Whether the raw materials needed to produce the fuel are readily available locally or regionally.

3. **Environmental Impact**

- Emissions profile (CO₂, NO_x, SO_x), biodegradability, toxicity, and overall ecological footprint compared to fossil fuels.

4. Production and Processing Cost

- Capital and operational costs, cost of technology, scalability, and long-term financial sustainability.

5. Infrastructure Requirements

- Need for new equipment, storage facilities, distribution networks, engines, or modifications to existing systems.

6. Technological Maturity

- Level of development, reliability, efficiency of existing technologies, and availability of technical expertise.

7. Energy Security Considerations

- Ability to reduce dependence on imported fuels and enhance local energy independence.

8. Socioeconomic Benefits

- Potential to create jobs, support farmers, strengthen rural economies, and diversify income sources.

9. Compatibility with Existing Systems

- Whether the fuel can be used in current engines, turbines, boilers, or requires system redesign.

10. Safety and Storage Properties

- Flammability, handling risks, storage stability, and transportation safety.

11. Government Policies and Incentives

- Availability of subsidies, tax incentives, regulations, and national energy strategies that favor certain fuels.

12. Sustainability and Long-Term Availability

- Whether the fuel source can be sustained without harming food security, land use, or ecosystems.

METHANOL PRODUCTION :

Methanol (CH_3OH), also known as methyl alcohol or wood alcohol, is one of the simplest alcohols and a versatile chemical with wide industrial and energy-related applications. It is a colorless, volatile liquid with a mild alcoholic odor and is fully miscible with water. Methanol is highly flammable and burns with a clean, almost invisible flame, making it a valuable fuel and chemical feedstock. Its molecular simplicity and reactivity allow it to participate in a broad range of chemical transformations, which underpins its importance in global chemical industries.

Historically, methanol was produced through the destructive distillation of wood, but modern large-scale production relies primarily on catalytic reactions involving natural gas—specifically, steam reforming of methane to produce synthesis gas (a mixture of CO and H_2), followed by catalytic conversion to methanol. Growing interest in renewable energy has also revived research into bio-methanol, which can be produced from biomass, municipal solid waste, agricultural residues, and even captured CO_2 combined with green hydrogen. These emerging pathways position methanol as a potential low-carbon fuel for the future.

Methanol's applications are broad and continue to expand. It serves as a key raw material for the production of formaldehyde, acetic acid, olefins, and various plastics, resins, and adhesives. In the energy sector, methanol is gaining attention as a clean alternative fuel. It can be blended with gasoline, used directly in spark-ignition engines, or converted into dimethyl ether (DME), a clean-burning diesel substitute. Methanol is also used in fuel cells, particularly in direct methanol fuel cells (DMFCs), which offer promising applications for portable power devices and small-scale electricity generation.

In addition, methanol has significance in marine transportation, where it is emerging as a low-emission fuel that reduces sulfur oxides (SO_x), nitrogen oxides (NO_x), and particulate matter compared to conventional marine fuels. Its relatively simple storage requirements and liquid form make it easier to handle than compressed or liquefied gaseous fuels. Economically, methanol is attractive due to its relatively low production cost, established global distribution network, and adaptability to various feedstocks.

Overall, methanol's versatility, cleaner combustion profile, and compatibility with renewable production pathways have made it a critical component of the global transition toward sustainable energy and cleaner fuels. Its role is expected to grow as industries and governments continue to seek environmentally friendly, cost-effective energy solutions.

METHANOL PRODUCTION :

1.Methanol Production Via Synthesis gas

Natural gas is the largest source of synthesis gas. Methane (CH_4) is the chief constituent of natural gas. Methanol is made from CH_4 in a series of three reactions.

Steam reforming reaction

In steam reforming, CH₄ reacts in a highly endothermic reaction with steam over a catalyst, typically based on nickel, at high temperatures (800–1000°C, 20–30 atm) to form CO and H₂.



Water–gas shift (WGS) reaction

A part of the CO formed reacts consequently with steam in the WGS reaction to yield more H₂ and also CO₂. The gas obtained is thus a mixture of H₂, CO, and CO₂. The compressed synthesis gas enters the converter containing copper zinc and catalyst and the methanol synthesis occurs according to the methanol synthesis reaction.



Conventional Methanol Production

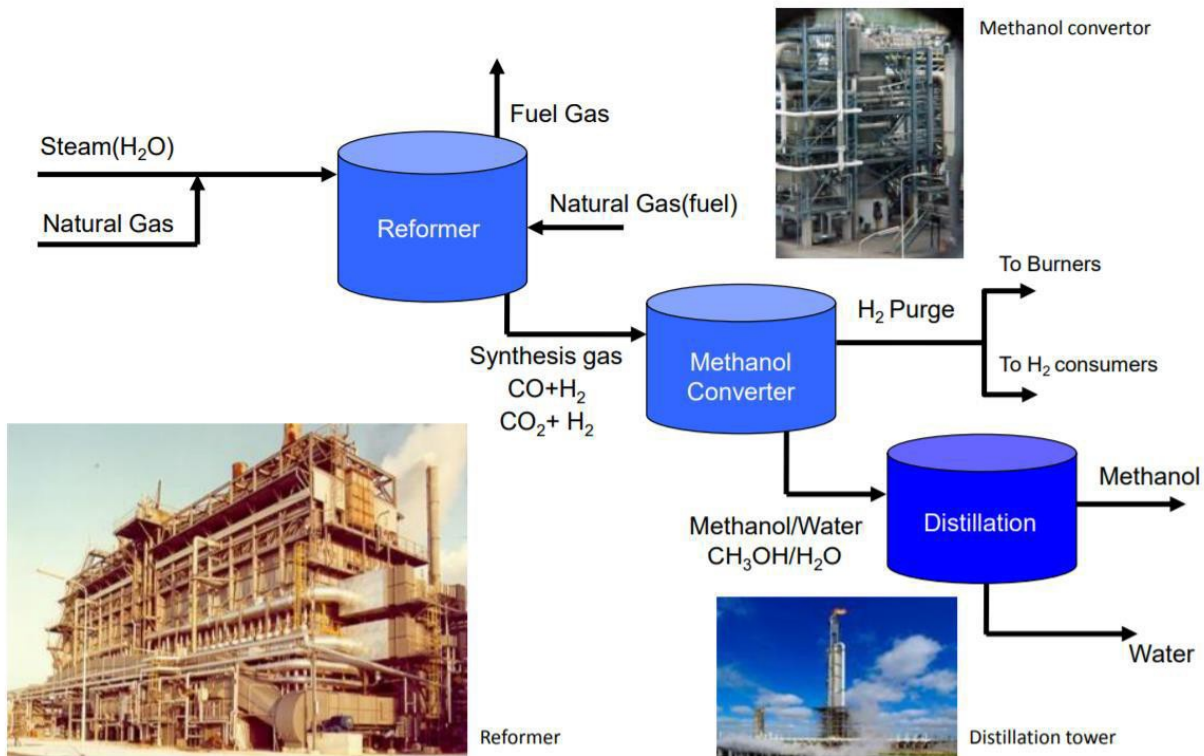
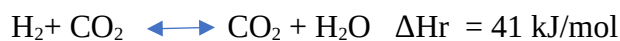


Fig 1.

Methanol synthesis reaction



If CO is used up in the methanol synthesis reaction the WGS reverses producing more CO.



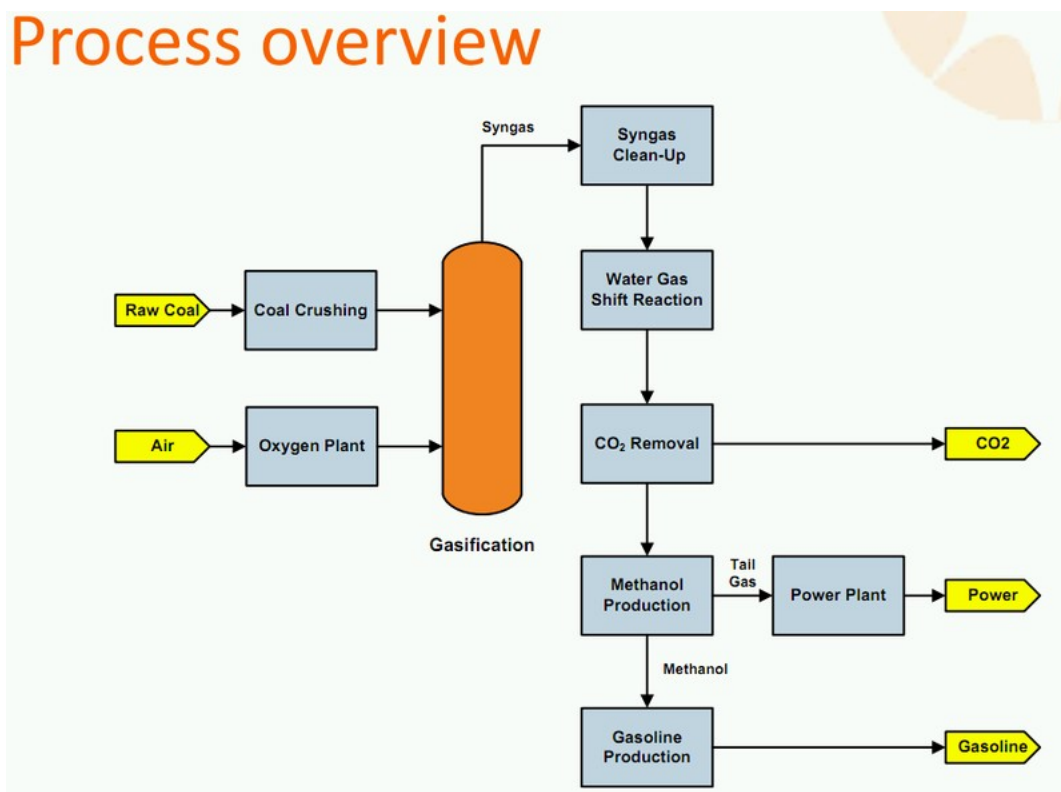
This reaction combines to produce approximately 40% conversion of CO to methanol. The

gas mixture is cooled and methanol and water condense out. The remaining gas is returned to the circulator, mixed with incoming compressed synthesis gas and recycled through the methanol converter.

Hence, the overall reactions by which methanol are produced from synthesis gas may be summarized into the following equation



METHANOL PRODUCTION BY COAL GASIFICATION



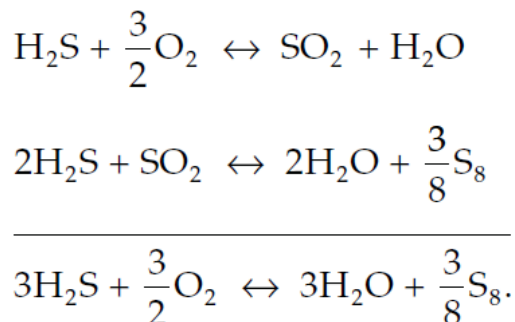
Methanol production from coal involves several steps that are

- Coal gasification
- Acid removal
- WGS reaction
- Methanol synthesis
- Methanol refining

The first step in the production of methanol is the gasification of coal to CO and H₂. The overall reaction for coal gasification is



The main by-product of coal gasification is acid gas, which is composed of hydrogen sulfide (H₂S) and CO₂. The H₂S needs to be removed from the synthesis gas. The first stage involves the combustion of one-third the total H₂S to form SO₂ and water; the second is a low temperature catalytic stage that involves the reaction of SO₂ with the remaining H₂S to ultimately yield water and elemental sulfur. The reaction is presented as follows:

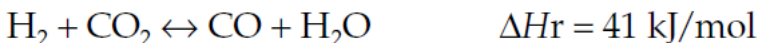


The following step of the WGS reaction usually utilizes a high-temperature shift reaction with Fe₃O₄ as a catalyst. The purpose of the WGS reaction is to adjust the ratio of H₂ to CO to 2:1.

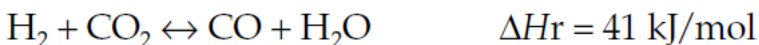
In the last step, CO and H₂ produce methanol by the methanol synthesis reaction. The most widely used catalyst is a mixture of copper, zinc oxide, and alumina, and we can use it to catalyze the production of methanol with better selectivity



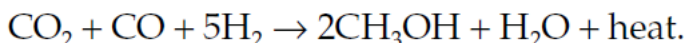
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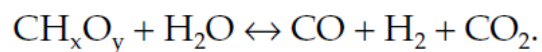


Biomass is an organic material, such as urban wood wastes, primary mill residues, forest residues, agricultural residues, and dedicated energy crops (e.g., sugar cane and sugar beets) that can be made into fuel.

Biomass can be converted to synthesis gas by a process called partial oxidation and later converted to methanol

Biomass $\rightarrow$ synthesis gas (CO , H_2) $\rightarrow$ CH_3OH

In the first step biomass undergoes gasification to produce synthesis gas. Carbon source is reacted with steam (or steam and oxygen) at very high temperatures to produce CO , and H_2 in the gasification is a process



The following equation shows that the synthesis gas is then reacted in the presence of a catalyst to yield CH_3OH by methanol synthesis reaction



BENEITS OF METHANOL FUEL

Clean Burning and Lower Emissions

Methanol burns more cleanly than gasoline or diesel, producing lower levels of nitrogen oxides (NO_x), sulfur oxides (SO_x), particulate matter, and carbon monoxide. This makes it attractive for reducing air pollution and improving urban air quality.

Lower Carbon Footprint

Methanol contains less carbon per unit of energy compared to fossil fuels. When produced from biomass, municipal waste, or captured CO_2 , methanol can significantly reduce greenhouse gas emissions.

High Octane Rating

Methanol has a high octane number, which enhances engine performance and efficiency. Engines running on methanol can achieve better power output and reduced knocking.

Ease of Storage and Transportation

Unlike hydrogen or liquefied natural gas (LNG), methanol is a liquid at ambient conditions, making it easier and safer to store, transport, and distribute using existing fuel infrastructure.

Versatile Feedstock Options

Methanol can be produced from natural gas, coal, biomass, agricultural residues, waste plastics, or even CO_2 combined with green hydrogen. This flexibility allows countries to use local resources and enhance energy security.

Cost-Effective Production

Methanol is relatively inexpensive to produce and has an established global market. It offers a low-cost alternative to petroleum-based fuels, especially in regions with abundant natural gas or biomass.

	Reduced	Engine	Wear
	Methanol combustion generates fewer deposits in engines, contributing to cleaner engine operation and potentially lower maintenance costs.		

Supports	Renewable	Energy	Integration
Renewable methanol (from biomass or CO ₂ + renewable hydrogen) enables storage of surplus renewable electricity in liquid form, supporting energy transition and decarbonization.			

	Safety	Advantages
	Methanol is less volatile and has a lower risk of explosion compared to gasoline. Although toxic when ingested, it is easier to contain in case of spills because it dissolves in water.	

Marine	and	Industrial	Applications
Methanol is emerging as a major low-emission fuel in shipping. It offers a practical pathway for the maritime industry to meet stricter environmental regulations without requiring complex cryogenic systems.			

Shortcomings/Limitations of Methanol Fuel

Lower Energy Density

Methanol contains about half the energy per liter of gasoline. This means vehicles require more fuel to travel the same distance, leading to reduced driving range and larger fuel tanks for equivalent performance.

Corrosiveness to Materials

Methanol can be corrosive to certain metals, plastics, and rubber components commonly used in fuel systems. Engines and storage systems may need special materials or modifications to avoid degradation.

Toxicity Concerns

Methanol is toxic if ingested and its vapors can be harmful if inhaled. Careful handling, storage, and safety protocols are required to avoid exposure.

Cold Start Challenges

Methanol has a lower vapor pressure and poor cold-start properties, making engines difficult to start in low-temperature conditions unless blended with other fuels or additives.

Water Absorption (Hygroscopic Nature)

Methanol readily absorbs water from the environment, which can dilute the fuel, reduce combustion efficiency, and cause corrosion in storage tanks and pipelines.

Infrastructure Limitations

Although easier to store than many alternative fuels, widespread use of methanol still requires upgrades to fuel systems, stations, and pipelines to accommodate its chemical properties.

Lower Lubricity

Methanol has poor lubricating properties compared to diesel or gasoline. This can increase wear on fuel pumps and injectors if not properly managed with additives.

Production Emissions (Depending on Feedstock)

When methanol is produced from coal or natural gas, it can generate significant CO₂ emissions. Only renewable methanol (from biomass or captured CO₂) offers major climate benefits.

Public Perception and Safety Misconceptions

Concerns over toxicity and the invisibility of methanol flames may affect consumer acceptance, despite the fuel being generally safer than gasoline in controlled conditions.

Economic Competitiveness in Some Regions

In places without available feedstock or established supply chains, methanol may be less economically attractive than other renewable fuels or conventional fuels.

Methanol: gasoline Blends

- Methanol may be used directly as an engine fuel just as LPG and ethanol. The direct use of methanol as an engine fuel has advantages and disadvantages.
- The relative very high-latent heat of vaporization of methanol results in a lower burning temperature in the engine cylinders as it does for ethanol.
- The direct use of methanol as an engine fuel in passenger vehicles would require nontrivial engine modifications and substantial changes in the lubrication system.
- Due to its high Reid Vapor Pressure (RVP) that is a measure of affected volatility of blended gasoline, use of methanol as gasoline blending is limited even though it has a high octane rating and is an excellent candidate of oxygenated hydrocarbons.
- When pure methanol is used, cold start problems can occur because it lacks the highly volatile compounds (butane, iso-butane, propane) that provide ignitable vapor to the engine even under the most frigid conditions.
- The addition of more volatile components to methanol is usually the preferred solution.

Methanol: Diesel Blends

- Methanol and diesel are not very miscible, which makes it difficult to mix the PM together as a diesel engine fuel.
- Methanol–diesel blends gave good performance when the amount of methanol in the mixed fuel does not exceed 30% by weight and combustion characteristics when the maximum methanol mass fraction was 20% by weight

- Methanol can be blended with diesel fuel although its corrosive nature creates a need for caution in the design of the engine and fuel system like in ethanol.
- Methanol has half the volumetric energy density relative to gasoline or diesel. It is usually mixed in percentages ranging from 5 to 85%.
- The combination rate of methanol in the fuel is 85%, which is called M85.
- At the same time, the aldehyde compound that comes along with burning methanol forms a strong acid. Therefore, the researchers should pay more attention to this fuel.

FUEL CELL

Introduction

Fuel cells are advanced energy conversion devices that generate electricity through electrochemical reactions rather than combustion, offering a highly efficient and environmentally friendly alternative to conventional power systems. At their core, fuel cells operate by combining a fuel, commonly hydrogen or methanol with oxygen from the air to produce electricity, water, and heat. Because this process does not involve burning fuel, fuel cells emit little to no pollutants, making them a promising technology for clean and sustainable energy generation.

In recent years, fuel cells have gained considerable attention due to their high efficiency, low emissions, and ability to provide continuous power as long as fuel is supplied. They are versatile systems that can be scaled from small portable devices and residential power units to large industrial applications, including transportation, backup power, and grid support. With increasing concerns about climate change, energy security, and the need for low-carbon technologies, fuel cells represent a significant step toward a more efficient and sustainable energy future. Advances in materials, catalysts, and hydrogen production technologies continue to improve their performance and affordability, positioning fuel cells as a key component of the emerging clean energy landscape.

Advantages of Fuel Cells Over Conventional Energy Sources

Higher Efficiency

Fuel cells convert chemical energy directly into electrical energy through electrochemical reactions, avoiding the energy losses associated with combustion. This allows them to achieve efficiencies significantly higher than traditional internal combustion engines and some thermal power plants

Low or Zero Emissions

When hydrogen is used as the primary fuel, the only by-products are water and heat. Even when using alternative fuels like methanol or natural gas, emissions of pollutants such as NO_x, SO_x, CO, and particulate matter remain far lower than those from conventional fossil fuel systems.

Quiet Operation

Fuel cells have few moving parts, resulting in much quieter operation compared to engines and turbines. This makes them ideal for residential, industrial, and mobile applications where noise reduction is important.

Scalability and Versatility

Fuel cells can be designed for a wide range of applications—from small portable electronics and backup power units to vehicles, ships, and large stationary power plants. Their modular design allows easy scaling to meet different energy demands.

Reliable and Continuous Power Supply

Unlike batteries, fuel cells can produce power continuously as long as fuel is available. This makes them highly reliable for critical applications such as hospitals, data centers, and remote installations.

Renewable Energy Integration

When hydrogen is produced from renewable sources such as solar, wind, or biomass, fuel cells become a completely clean energy solution. They also support energy storage by converting excess renewable electricity into hydrogen for later use.

Reduced Dependence on Fossil Fuels

Fuel cells provide an alternative to oil- and coal-based energy systems, helping countries enhance energy security, diversify energy supply, and reduce exposure to fluctuating fuel prices.

Minimal Environmental Impact

Fuel cells have lower environmental footprints throughout their life cycle with less noise, fewer emissions, and reduced thermal pollution when compared to traditional power systems.

Fuel cells generate electricity through an electrochemical reaction rather than combustion. The basic working principle involves converting the chemical energy of a fuel—usually hydrogen—directly into electrical energy, water, and heat.

A typical fuel cell consists of three main components:

anode, cathode, and electrolyte.

Hydrogen Fuel Enters the Anode

At the anode, hydrogen molecules (H_2) are supplied and come into contact with a catalyst, often platinum. The catalyst facilitates the splitting of hydrogen into protons (H^+) and electrons (e^-).

Separation of Charges

The electrolyte allows only the protons (H^+ ions) to pass through to the cathode.

The electrons cannot pass through the electrolyte and are forced to travel through an external circuit, generating electric current that can be used to do work.

Oxygen Enters the Cathode

Oxygen (from air) is supplied to the cathode. Here, the oxygen molecules combine with the protons arriving through the electrolyte and the electrons arriving through the external circuit.

Formation of Water and Heat

At the cathode, the reaction produces water (H_2O) and releases heat as by-products.

Types of Fuel Cells

a. Proton Exchange Membrane Fuel Cell (PEMFC)

- Uses a solid polymer electrolyte.
- Operates at low temperatures (60–80°C).
- Fast start-up, high power density.
- Common in vehicles, portable devices, and residential power.

b. Alkaline Fuel Cell (AFC)

- Uses an alkaline electrolyte (e.g., potassium hydroxide).
- High efficiency and good performance at low temperatures.
- Used historically in spacecraft (e.g., NASA missions).

c. Solid Oxide Fuel Cell (SOFC)

- Utilizes a solid ceramic electrolyte.
- Operates at very high temperatures (600–1,000°C).
- Excellent efficiency; can use multiple fuels (hydrogen, natural gas).
- Used in stationary power plants and industrial systems.

d. Molten Carbonate Fuel Cell (MCFC)

- Uses molten carbonate salts as the electrolyte.
- High-temperature operation (600–700°C).
- Suitable for large-scale stationary power generation.

e. Phosphoric Acid Fuel Cell (PAFC)

- Uses liquid phosphoric acid electrolyte.
- Moderate temperatures (150–200°C).
- Often used in commercial buildings for combined heat and power (CHP).

f. Direct Methanol Fuel Cell (DMFC)

- Uses methanol directly as fuel without external reforming.
- Low temperature, compact, suitable for portable electronics and small generators.